Lec 8. Interval Estimation

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Learning objectives

After this section, you should be able to ...

- understand the confidence interval
- find the CI

Statistical Estimation

- **Point Estimation:** using the data to calculate a single estimate of the parameter of interest. For example, we often use the sample mean $\bar{x}$ to estimate the population mean μ.

- **Interval Estimation:** provides a range of values (an interval) that may contain the unknown parameter (such as the population mean μ).

Motivation

Table 8.1: Dyken and White (1959, JAMA)'s Trial of Cortisone in Stroke: Effect on Death

	Cortisone, n = 17	Placebo, n = 19	Relative Risk, 95% CI*
Died	13(76.5%)	10(52.6%)	1.45(0.88 – 2.40)

*CI indicates confidence interval.

- Why we need CI?
- With CI, we have additional information about the likely values we are trying to estimate.

Interval Estimation

Definition

An interval estimate of a real-values parameter θ is any pair of functions, $L(x_1, \ldots, x_n)$ and $U(x_1, \ldots, x_n)$, of a sample that satisfy $L(\boldsymbol{x}) \leq U(\boldsymbol{x})$ for all $\boldsymbol{x} \in \mathcal{X}$. If $\boldsymbol{X} = \boldsymbol{x}$ is observed, the inference $L(\boldsymbol{x}) \leq \theta \leq U(\boldsymbol{x})$ is made. The random interval $[L(\boldsymbol{X}), U(\boldsymbol{X})]$ is called an interval estimator.

- **Coverage probability**:

$$P_\theta(\theta \in [L(\boldsymbol{X}), U(\boldsymbol{X})])$$

- **Precision**: the size of interval, e.g., $E[U(\boldsymbol{X}) - L(\boldsymbol{X})]$

8.1 Confidence Interval

Definition

Let $\mathbf{X} = (X_1, \ldots, X_n)$ be a sample from population $\{f(x \mid \theta), \theta \in \Theta\}$ with support $\mathcal{X}$. Let $L_n(\boldsymbol{X})$ and $U_n(\boldsymbol{X})$ be statistics such that $L_n(x) \leq U_n(x), \forall x \in \mathcal{X}$. Then $[L_n(\boldsymbol{X}), U_n(\boldsymbol{X})]$ is a $100(1-\alpha)\%$ confidence interval (CI) for θ if

$$P_\theta\left(L_n(\boldsymbol{X}) \leq \theta \leq U_n(\boldsymbol{X})\right) \geq 1 - \alpha$$

for all $\theta \in \Theta$. The interval $(L_n(\boldsymbol{x}), U_n(\boldsymbol{x}))$ is a large sample (asymptotic) $100(1-\alpha)\%$ CI for θ if

$$P_\theta\left(L_n(\boldsymbol{X}) < \theta < U_n(\boldsymbol{X})\right) \to 1 - \alpha$$

for all $\theta \in \Theta$ as $n \to \infty$. α is called confidence level.

Remark:

- It will sometimes be more natural to use an open interval $(L_n(x), U_n(x))$ or even a half-open and half-closed interval
- The interval can also be one-sided interval: $(-\infty, U_n(X)]$ or $[L_n(X), \infty)$
- This definition can be extended to multi-dimensional cases: 2D, confidence band; $\geq$ 3D, confidence region.
- Naturally, we want our set to have **small size** and **large coverage probability**, but such sets are usually difficult to construct.
- A typical CI form of θ is $\hat{\theta} \pm sd \times F_q$, where F_q is the quantile and sd is the standard error of $\hat{\theta}$. The value $sd \times F_q$ is called the margin of error

Find the CI

8.1.1 The exact CI

Definition

The random variable $g(\boldsymbol{X}, \theta)$ is a pivot or pivotal quantity (枢轴变量) if its distribution is independent θ.

- For pivotal quantity $g(\boldsymbol{X}, \theta)$, find $a < b$ such that

$$P_\theta(a \leq g(\boldsymbol{X}, \theta) \leq b) \geq 1 - \alpha$$

- Re-write the interval as

$$C(\boldsymbol{x}) = \{\theta : a \leq g(\boldsymbol{x}, \theta) \leq b\} = \{\theta : L_n(\boldsymbol{x}, a) \leq \theta \leq U_n(\boldsymbol{x}, b)\}$$

and $C(\boldsymbol{X})$ is a $1 - \alpha$ confidence set for θ.

Examples

- CIs for the mean and variance in normal population
- CIs for the difference of means and the ratio of variances in two-sample normal populations
- CI for the mean of Exponential distribution

Find the CI

8.1.2 The asymptotic CI

Definition

The quantity $g(\boldsymbol{X}, \theta)$ is an asymptotic pivot if its limiting distribution is independent of θ.

A large sample $100(1-\alpha)\%$ CI for θ can be easily obtained.

CIs for the proportion: we denote $\kappa \equiv z_{\alpha/2} = \Phi^{-1}(1-\alpha/2), \hat{p} = X/n$ and $\hat{q} = 1 - \hat{p}$.

- Wilson interval: $CI_W = \frac{X+\kappa^2/2}{n+\kappa^2} \pm \frac{\kappa n^{1/2}}{n+\kappa^2} \left(\hat{p}\hat{q} + \kappa^2/(4n)\right)^{1/2}$
- Wald interval: $CI_s = \hat{p} \pm \kappa n^{-1/2} (\hat{p}\hat{q})^{1/2}$
- Agresti-Coull interval: $CI_{AC} = \tilde{p} \pm \kappa (\tilde{p}\tilde{q})^{1/2} \tilde{n}^{-1/2}$, where $\tilde{X} = X + \kappa^2/2$ and $\tilde{n} = n + \kappa^2$. Let $\tilde{p} = \widetilde{X}/\tilde{n}$ and $\tilde{q} = 1 - \tilde{p}$.
- Exact interval: $CI_{CP} = [L_{CP}(x), U_{CP}(x)]$, where $L_{CP}(x)$ and $U_{CP}(x)$ are, respectively, the solutions in p to the equations $P_p(X \geq x) = \alpha/2$ and $P_p(X \leq x) = \alpha/2$.

8.1.3 Bootstrap Confidence Interval

- In 1979 , Brad Efron invented a revolutionary new statistical procedure called the **bootstrap**(自助法).

- The bootstrap idea is illustrated in the diagram below. The propagation of the chance variation in the initial sample into the estimator $\hat{\theta}$ is approximated by the observed impact of chance variation in the resampled data (with replacement) upon the bootstrap estimates.

$$
\begin{array}{ccccccccc}
f & \longrightarrow & \begin{bmatrix} X_1 \\ X_2 \\ \vdots \\ X_n \end{bmatrix} & \Longrightarrow & \hat{f}_n & \longrightarrow & \begin{bmatrix} X^*_{1(\mathbf{1})} \\ X^*_{2(\mathbf{1})} \\ \vdots \\ X^*_{n(\mathbf{1})} \end{bmatrix}, & \cdots, & \begin{bmatrix} X^*_{1(M)} \\ X^*_{2(M)} \\ \vdots \\ X^*_{n(\mathbf{M})} \end{bmatrix} \\
 & & \downarrow & & & & \downarrow & & \downarrow \\
 & & \hat{\theta} & & & & \hat{\theta}^*_{\mathbf{1}}, & \cdots, & \hat{\theta}^*_M
\end{array}
$$

- Since $\hat{f}_n$ approximates f, the unknown distribution of $\hat{\theta}$, based on f, is approximately the distribution of $\hat{\theta}^*_1, \ldots, \hat{\theta}^*_M$, based on $\hat{f}_n$. That approximating distribution is used to set confidence intervals.

- **percentile confidence interval**:

$$L = \hat{\theta}^*_{(0.025)} \quad U = \hat{\theta}^*_{(0.975)}$$

But this does not take full account of the difference between θ for f and $\hat{\theta}$, the true value for $\hat{f}_n$. We can do a bit better.

- **The basic bootstrap confidence interval**: the behavior of $\theta - \hat{\theta}$ is approximately the same as the behavior of $\hat{\theta} - \hat{\theta}^*$. Thus

$$\begin{aligned}
0.95 &\approx \mathbf{P}\left[\hat{\theta}^*_{(.025)} \le \hat{\theta}^* \le \hat{\theta}^*_{(0.975)}\right] \\
&= \mathbf{P}\left[\hat{\theta}^*_{(0.025)} - \hat{\theta} \le \hat{\theta}^* - \hat{\theta} \le \hat{\theta}^*_{(0.975)} - \hat{\theta}\right] \\
&= \mathbf{P}\left[\hat{\theta} - \hat{\theta}^*_{(0.025)} \ge \hat{\theta} - \hat{\theta}^* \ge \hat{\theta} - \hat{\theta}^*_{(0.975)}\right] \\
&\approx \mathbf{P}\left[\hat{\theta} - \hat{\theta}^*_{(0.025)} \ge \theta - \hat{\theta} \ge \hat{\theta} - \hat{\theta}^*_{(0.975)}\right] \\
&= \mathbf{P}\left[2\hat{\theta} - \hat{\theta}^*_{0.025} \ge \theta \ge 2\hat{\theta} - \hat{\theta}^*_{(0.975)}\right]
\end{aligned}$$

so

$$L = 2\hat{\theta} - \hat{\theta}^*_{(0.975)} \quad U = 2\hat{\theta} - \hat{\theta}^*_{(0.025)}$$

8.1.4 Other methods

We will discuss the following methods later.

- Inverting a test Statistic
- Bayesian interval